1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlights that immune-contexture metrics (Immunoscore/immunotherapy biomarkers) often predict outcome in MIBC, yet microenvironment phenotypes like SARIFA (adipocyte–CAF interactions) associate with metastasis and may modulate immune activity. The puzzle is: why do some TIL-rich tumors fail to translate immune infiltration into durable immunotherapy benefit? 2) Tier 1 paths used: I built on the Immunotherapy biomarker and Immunoscore paths because they establish that T cell infiltration and spatial immune architecture are expected to predict IO benefit; I also used the SARIFA path because adipocyte–CAF programs and associated stromal remodeling (linked mechanistically to TGF-β) provide a plausible axis that could suppress or spatially exclude functional T cells. These three together create a mechanistic hypothesis that stromal/adipocyte-driven TGF-β signaling undermines the functional benefit of TILs. 3) Tier 2 support: MDK, PTGDS and EV-mRNA signals support noncanonical stromal/tumor signaling and systemic biomarker readouts; methylation markers (SDC2/TFPI2) and metabolic markers indicate that epigenetic and systemic context may further stratify tumors where stromal programs dominate. These support testing combined tumor/stroma molecular readouts with imaging. 4) Why therapy_response endpoint: The Ontologist emphasis is on immunotherapy biomarkers and immunoscore applicability; the translational question of greatest clinical impact is predicting which patients will respond to checkpoint blockade (therapy response proxy). This endpoint directly tests the mechanism (immune infiltration + stromal suppression → IO resistance). 5) Deliberate exclusions: I excluded using multiple immune markers (CD3/CD8 separately), full ECM/CAF gene lists, and broad metabolic panels from the core hypothesis to keep it focused (max 2 anchor genes). Those are placed in secondary_modifiers for follow-up analyses?

Working Hypothesis

In muscle-invasive bladder cancer (MIBC), high tumor-infiltrating lymphocyte (TIL) density is necessary but not sufficient for response to PD-1/PD-L1 blockade. A SARIFA-like adipocyte–CAF program drives elevated TGFB1 signaling and PD-L1 (CD274) expression, causing spatial exclusion and functional suppression of CD8+ T cells within the tumor microenvironment. Consequently, tumors with high TIL density but active TGFB1 signaling and high CD274 will exhibit poor IO response. The hypothesis predicts that a combined readout of TIL density and TGFB1/CD274 status will more accurately predict immunotherapy nonresponse than TIL alone. Moreover, in tumors with this stromal program, inhibition of TGF-β signaling will restore T cell penetration and cytotoxicity and enhance the efficacy of PD-1/PD-L1 blockade. Secondary modifiers (MDK, PTGDS, SDC2, TFPI2) and EV-mRNA/metabolic methylation signatures will further discriminate IO responders from nonresponders within the high-TIL cohort and may guide stratified therapy decisions.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	TGFB1, CD274
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGFB1: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

CD274: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	MDK	Supporting modifier from knowledge graph
Secondary	PTGDS	Supporting modifier from knowledge graph
Secondary	SDC2	Supporting modifier from knowledge graph
Secondary	TFPI2	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: , PD-L1, and ECM remodeling markers. Endpoints include objective response rate, PFS/OS, TIL localization metrics, and pathway activity scores.

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- MDK: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PTGDS: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.